

Dividing Large Numbers

Year 6 reference sheet - short division, factors, long division & remainders

1. The key words

$$\begin{array}{ccc} 4,536 & \div & 7 = 648 \\ \text{dividend} & & \text{divisor} \quad \text{quotient} \end{array}$$

The **dividend** is the number being shared or grouped. The **divisor** is what you divide by. The **quotient** is the answer. Anything left over is the **remainder**.

Always estimate first: $6,182 \div 29$ is about $6,000 \div 30 = 200$. **Always check after:** multiply your answer by the divisor (then add any remainder) and you should get the dividend back.

2. Short division ("bus stop") - you know this from Year 5

$$\begin{array}{r} 6 \ 4 \ 8 \\ 7 \overline{) 4 \ 5 \ 3 \ 6} \end{array}$$

small orange numbers = carried remainders

Work from **left to right**, one digit at a time:

- $4 \div 7 = 0$, so carry the 4 to make **45**
- $45 \div 7 = 6 \text{ r } 3$. Write 6 on top, carry the 3
- $33 \div 7 = 4 \text{ r } 5$. Write 4, carry the 5
- $56 \div 7 = 8$. Answer: **648**

3. Dividing by a 2-digit number: write the multiples first

Multiples of 24

- $1 \times 24 = 24$
- $2 \times 24 = 48$
- $3 \times 24 = 72$
- $4 \times 24 = 96$
- $5 \times 24 = 120$
- $6 \times 24 = 144$
- $7 \times 24 = 168$
- $8 \times 24 = 192$
- $9 \times 24 = 216$

$$\begin{array}{r} 3 \ 0 \ 2 \\ 24 \overline{) 7 \ 2 \ 4 \ 8} \end{array}$$

$$7,248 \div 24$$

- $7 \div 24 = 0$, so use **72**: $72 \div 24 = 3$
- $4 \div 24 = 0 \text{ r } 4$. Write the 0!
- $48 \div 24 = 2$

Answer: **302**. Check: $302 \times 24 = 7,248$

4. Division using factors (split the divisor into two easy steps)

If the divisor has a factor pair you know well, divide by one factor, then the other.

$$2,688 \div 24 \quad 24 = 4 \times 6, \text{ so } \div 4 \text{ then } \div 6$$

$$2,688 \div 4 = 672$$

$$672 \div 6 = 112$$

$$\text{So } 2,688 \div 24 = 112$$

Good factor pairs: $12 = 3 \times 4$, $15 = 3 \times 5$, $16 = 4 \times 4$, $18 = 2 \times 9$, $24 = 4 \times 6$, $25 = 5 \times 5$, $36 = 4 \times 9$.

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Long division & remainders

5. Long division: Divide, Multiply, Subtract, Bring down

$$\begin{array}{r} 364 \\ 24 \overline{) 8736} \\ \underline{- 72} \\ 153 \\ \underline{- 144} \\ 96 \\ \underline{- 96} \\ 0 \end{array}$$

$8,736 \div 24$ (written out in full, so nothing is kept in your head)

- **Divide:** $8 \div 24$ won't go, so use 87. $87 \div 24 = 3$ ($3 \times 24 = 72$)
- **Multiply & Subtract:** $87 - 72 = 15$
- **Bring down** the next digit (3) to make **153**
- $153 \div 24 = 6$ ($6 \times 24 = 144$). $153 - 144 = 9$
- Bring down the 6 to make **96**. $96 \div 24 = 4$, remainder 0
- Answer: **364**. Check: $364 \times 24 = 8,736$

Remember it with: **Does McDonald's Sell Burgers?** (Divide, Multiply, Subtract, Bring down - then repeat.)

6. Remainders: four ways to write them

$3,754 \div 16 = 234$ with **10** left over. Depending on the question, write it as:

As a remainder $234 \text{ r } 10$

As a fraction $234 \frac{10}{16} = 234 \frac{5}{8}$ (the remainder goes over the divisor, then simplify)

As a decimal 234.625 ($10 \div 16 = 0.625$)

Rounded see the stories below

7. Remainders in real life: read the question!

Round UP when everyone or everything must fit.
437 people, coaches seat 52: $437 \div 52 = 8 \text{ r } 21$,
so you need **9 coaches**.

Round DOWN when only full groups count. 1,000
eggs in boxes of 12: $1,000 \div 12 = 83 \text{ r } 4$, so **83 full boxes**.

8. Top tips

- Write the **multiples of the divisor** down the side before you start. It saves lots of time.
- If a step won't go, write a **0** in the answer ($7,248 \div 24 = 302$, not 32).
- Your remainder must always be **smaller than the divisor**. If not, the digit above can go up by 1.
- Use **squared paper** so each digit has its own box and the columns line up.
- Short division and long division give the **same answer**. Long division just writes the subtractions down.